



INVESTIGATION OF CORRELATION BETWEEN GEOMAGNETIC-AND-SOLAR INDICES AND GROUND-BASED NEUTRON MONITOR COUNT RATE

Natdaporn Thongphan*, Kledsai Poopakun, Kamolporn Kanthawong, Pranee Disrattakit
Mahidol Wittayanusorn School, Salaya, Nakhon Pathom, Thailand 73170

*e-mail: natdaporn.tho_g33@mwit.ac.th

Abstract:

Galactic cosmic rays (GCRs) are highly energetic particles originating outside our solar system. As they travel through the heliosphere, their intensity and energy are modulated by the solar activity, a phenomenon known as solar modulation. This modulation can be monitored via the neutron count rate (NCR) measured by ground-based neutron monitors. In this study, we examine the relationship between NCR and solar and geomagnetic indices, including sunspot number (SSN), F10.7 solar radio flux, planetary Kp index, and auroral electrojet (AE) index. These indices represent different aspects of solar and geomagnetic activity. To investigate these relationships, we collected, visualized, and analyzed data using Python with correlation and time series comparisons. Our results indicate a negative correlation between NCR and the aurora indices AE and Kp, suggesting that as auroral activity increases, cosmic ray levels decrease. Thus, geomagnetic storms, as indicated by these indices, appear to play a significant role in modulating cosmic ray levels. Given that aurora occur during geomagnetic storms, which can disturb radio communication, satellite operations, and GPS accuracy on Earth. So, identifying the best predictive indices is important. Ultimately, understanding which indices most closely match cosmic ray behavior may improve space weather forecasting.